Solution Requirements

Multimodal Attendance & Engagement Agent • Nirbhay Dubey | B.Tech CSE (Year 1)

1. Purpose

This document lists what the Multimodal Attendance & Engagement Agent must do (functional requirements) and the qualities it must have (non-functional requirements). Each requirement has an ID so it can be traced to testing.

2. Functional Requirements

ID	Requirement	Priority
FR1	Capture video from the webcam and fall back to a synthetic classroom video when no camera exists.	Must
FR2	Detect faces in each frame and track them with persistent track IDs.	Must
FR3	Enroll a student only with explicit consent, storing a face embedding and photo locally.	Must
FR4	Recognize enrolled students; show unknown faces as anonymous "Student #ID".	Must
FR5	Estimate 3D head pose (yaw, pitch, roll) and eye gaze, including blink detection.	Must
FR6	Extract audio features and classify the room as QUIET, SPEECH, MODERATE_NOISE, or DISTURBANCE.	Must
FR7	Synchronize video observations with audio chunks in a time-aligned buffer.	Must
FR8	Compute a composite engagement score per student and for the class, with HIGH / MEDIUM / LOW tiers.	Must
FR9	Start and end lecture sessions; accumulate presence time and classify PRESENT, PARTIAL, or ABSENT.	Must
FR10	Allow the lecturer to override a status with a stored reason.	Should
FR11	Generate rule-based teaching recommendations with priorities and a 30 s debounce.	Should
FR12	Export attendance for a session as CSV and browse past sessions.	Should
FR13	Stream live telemetry over WebSocket and expose REST endpoints for the dashboard.	Should
FR14	Provide an 8-page dashboard: Overview, Live Monitoring, Attendance, Analytics, Suggestions, Reports, Exports, Settings.	Must
FR15	Let the lecturer pause the camera and adjust attendance thresholds and refresh interval.	Could
FR16	Deregister a student and purge their embedding, photo, and login; delete a course with cascade cleanup.	Should
FR17	Restrict joining and attendance qualification to students enrolled in the course.	Should

3. Non-Functional Requirements

ID	Category	Requirement
NFR1	Privacy	All processing and storage are local. No video, audio, or biometric data is sent to the cloud.
NFR2	Privacy	No raw video is written to disk unless an administrator enables it.
NFR3	Performance	Live video should run at roughly 25-30 fps on a standard laptop; telemetry is broadcast at 4 Hz.
NFR4	Reliability	If audio is missing, the system defaults to a QUIET state and keeps running.
NFR5	Reliability	If the landmark model is unavailable, pose falls back to a bounding-box estimate.
NFR6	Usability	Lecturer can run a full session using the dashboard only, with no command-line steps after startup.
NFR7	Portability	Runs on macOS, Linux, and Windows with Python 3.11.
NFR8	Maintainability	Services are separated by module; automated tests exist for each service (12 suites).
NFR9	Fairness	EMA smoothing prevents blinks or brief glances from causing abrupt score drops.
NFR10	Auditability	Manual changes to attendance carry a reason.

4. Constraints and Assumptions

- A single camera covers only a limited field of view; large halls need wide-angle or multiple cameras (future work).
- Engagement is a heuristic index of gaze, pose stability, and acoustics. It does not measure understanding or emotion.
- Microphone audio must be posted to `/api/audio/chunk`; there is no built-in background microphone listener.
- The recognition threshold (cosine distance 0.15) is conservative, so poor lighting can cause an enrolled student to appear as anonymous.

5. Traceability

Requirements map to test suites: FR1-FR2 to *test_face_detection*, FR3-FR4 to *test_face_recognition*, FR5 to *test_head_pose_gaze*, FR6 to *test_audio_analysis* and *test_vad_noise_classifier*, FR7 to *test_multimodal_alignment*, FR8 to *test_engagement_engine*, FR9-FR10 to *test_attendance_service*, FR11 to *test_reasoning_engine*, and FR12-FR13 to *test_api_endpoints* and *test_end_to_end_pipeline*.